

7/24/23, 2:53 PM

CodioProjectFileEditFindViewToolsEducationHelpRunProject Index (static)Debug

built_in.pyTerminalx

Welcome to Ubuntu 18.04.4 LTS (GNU/Linux 5.19.0-1028-aws x86_64)

* Documentation: https://help.ubuntu.com

* Management: https://landscape.canonical.com

* Support: https://ubuntu.com/advantage

*

* Welcome to the Codio Terminal!

*

* https://docs.codio.com/develop/develop/ide/boxes/overview

*

* Your Codio Box domain is: clockstatus-oceangreen.codio.io

*

Last login: Mon Jul 24 08:44:17 2023 from 192.168.10.226

codio@clockstatus-oceangreen:~/workspace\$ python3 code/built_in.py

codio@clockstatus-oceangreen:~/workspace\$ python3 code/built_in.py

Please wait a moment while I gather a list of all available modules...

/usr/lib/python3/dist-packages/IPython/kernel/_init_.py:13: ShimWarning: The `IPython.kernel` package has been deprecated since

hon 4.0.You should import from ipykernel or jupyter_client instead.

"You should import from ipykernel or jupyter_client instead.", ShimWarning)

DEBUG:pip.vcs:Registered VCS backend: git

DEBUG:pip.vcs:Registered VCS backend: hg

DEBUG:pip.vcs:Registered VCS backend: svn

DEBUG:pip.vcs:Registered VCS backend: bzr

Crypto	argparse	importlib	secretstorage
IPython	array	initializing	select
__future__	asgiref	inspect	selectors
__ast	asn1crypto	io	setuptools
asyncio	ast	ipaddress	shelve
bisect	asynchat	ipython_genutils	shlex
blake2	asyncio	itertools	shutil
bootlocale	asyncore	json	signal
bz2	atexit	keyring	simplegeneric
cffi_backend	audioop	keyword	site
codecs	autoreload	lib2to3	sitcustomize
codecs_cn	base64	linecache	six
codecs_hk	bdb	locale	smtpd
codecs_iso2022	binascii	lockfile	smtplib
codecs_jp	binhex	logging	sndhdr
codecs_kr	bisect	lsb_release	socket
codecs_tw	built_in	lzma	socketserver
collections	builtins	macpath	softwareproperties
collections_abc	bz2	macurl2path	spwd
compat_pickle	cProfile	mailbox	sqlite3
compression	calendar	mailcap	sqlparse
crypt	certifi	marshal	sre_compile
csv	cgi	math	sre_constants
ctypes	cgitb	mimetypes	sre_parse
ctypes_test	chardet	mmap	ssh_import_id
curses	chunk	module_review	ssl
curses_panel	cmath	modulefinder	stat
datetime	cmd	multiply	statistics
dbm	code	multiprocessing	storemagic
dbus_bindings	codecs	netifaces	string

Guide x

Collapse

Built-In Packages

Standard Library

While you can write your own packages, there are built-in packages you can use as well. Every programming language has something called a standard library, which is a collection of many packages to help the end user more effectively write code.

You know that you are using the standard library when you import a package without having to download the package first. For example, the `math` package is already installed on your system as it is a part of the standard library. Print the value of Pi.

```
print(pi)
```

TRY IT

Command was successfully executed.

You get an error because `pi` is not defined. While the `math` package is already installed on your system, you need to import it from the standard library before you can use it.

```
import math

print(math.pi)
```

TRY IT

Try these variations:

- Print the contents of the `math` package.

```
import math

print(dir(math))
```

- Print all of the modules in Python's standard library.

```
print(help('modules'))
```

▼ What is happening?

The output from the code above is wider than the output area, so you do not see nicely formatted columns for all of the modules. The output should be more legible if you run it in the terminal.

RUN IN TERMINAL

TRY IT

Command was successfully executed.

Power of the Standard Library

Python is considered to be a “batteries included” language, which means their standard library is quite large. Without having to install a third-party package, you can do the following in Python:

- Parse strings (`re` and `string`)
- Perform cryptographic tasks (`hashlib` and `secrets`)
- Work with other file formats (`csv` and `json`)
- Create GUIs (`tkinter`)
- Communicate with the OS (`os` and `io`)

Python has a large standard library, which means you can perform many tasks without having to install any third-party packages.

Try these variations:

- Load and print a JSON file.

```
import json

with open('code/star_wars.json') as data:
    luke = json.load(data)

output = json.dumps(luke, indent=2)
print(output)
```

- Encrypt a string.

```
import hashlib

secret = 'Shhhh... this is very important'
byte_string = secret.encode()
hashed_secret = hashlib.sha256(byte_string)

print(hashed_secret.hexdigest())
```

TRY IT

Command was successfully executed.

Reading Question

How do you use a package from the standard library in Python?

☐ Download then install the package.

☐ Download the package first.

☐ Nothing, just start using the package.

☒ Import the package first.

https://codio.com/sandipan-dey/modules-and-packages/terminal/backend-guide

1/1